Simulation Experiment 03

Code for MATLAB:

```
clc;
clear;
close all;

% Input parameters
Fs = 1000;       % Sampling frequency (Hz)
Fc = 100;        % Cutoff frequency (Hz)
N = 3;           % Filter order

% Analog Butterworth filter
Wc = 2*pi*Fc;
[ba, aa] = butter(N, Wc, 's');

% 1. Impulse Invariance Method
[b1, a1] = impinvar(ba, aa, Fs);

% 2. Bilinear Transformation without Prewarping
[b2, a2] = bilinear(ba, aa, Fs);

% 3. Bilinear Transformation with Prewarping
Wc_pre = 2*Fs*tan(pi*Fc/Fs);
[ba_pre, aa_pre] = butter(N, Wc_pre, 's');
[b3, a3] = bilinear(ba_pre, aa_pre, Fs);

% Display coefficients
fprintf('IMPULSE INVARIANCE METHOD\n');
disp('Numerator:'); disp(b1);
disp('Denominator:'); disp(a1);

fprintf('BILINEAR TRANSFORMATION\n');
disp('Numerator:'); disp(b2);
disp('Denominator:'); disp(a2);

fprintf('BILINEAR WITH PREWARPING\n');
disp('Numerator:'); disp(b3);
disp('Denominator:'); disp(a3);

% Frequency responses
[H1, f] = freqz(b1, a1, 2048, Fs);
H2 = freqz(b2, a2, 2048, Fs);
H3 = freqz(b3, a3, 2048, Fs);

% Magnitude response comparison
figure;
plot(f, 20*log10(max(abs(H1),1e-10)), 'b', ...
     f, 20*log10(max(abs(H2),1e-10)), 'r--', ...
     f, 20*log10(max(abs(H3),1e-10)), 'g-.');
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('IIR Filter Magnitude Response Comparison');
legend('Impulse Invariance', ...
       'Bilinear', 'Bilinear with Prewarping');
xline(Fc, ':k', 'Cutoff');

% Phase response comparison
figure;
```

```
plot(f, unwrap(angle(H1))*180/pi, 'b', ...
     f, unwrap(angle(H2))*180/pi, 'r--', ...
     f, unwrap(angle(H3))*180/pi, 'g-.');
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('IIR Filter Phase Response Comparison');
legend('Impulse Invariance', ...
       'Bilinear', 'Bilinear with Prewarping');

fprintf('\nFilter Order: %d\n', N);
fprintf('Sampling Frequency: %d Hz\n', Fs);
fprintf('Desired Cutoff Frequency: %d Hz\n', Fc);
fprintf('Prewarped Analog Cutoff: %.2f rad/s\n', Wc_pre);
```

Command window results:

1. Impulse Invariance Method
   - Numerator coefficients: `[0, 0.0797, 0.0526, 0]`
   - Denominator coefficients: `[1, -1.7833, 1.2003, -0.2846]`

2. Bilinear Transformation Method
   - Numerator coefficients: `[0.0167, 0.0501, 0.0501, 0.0167]`
   - Denominator coefficients: `[1, -1.7977, 1.2211, -0.2898]`

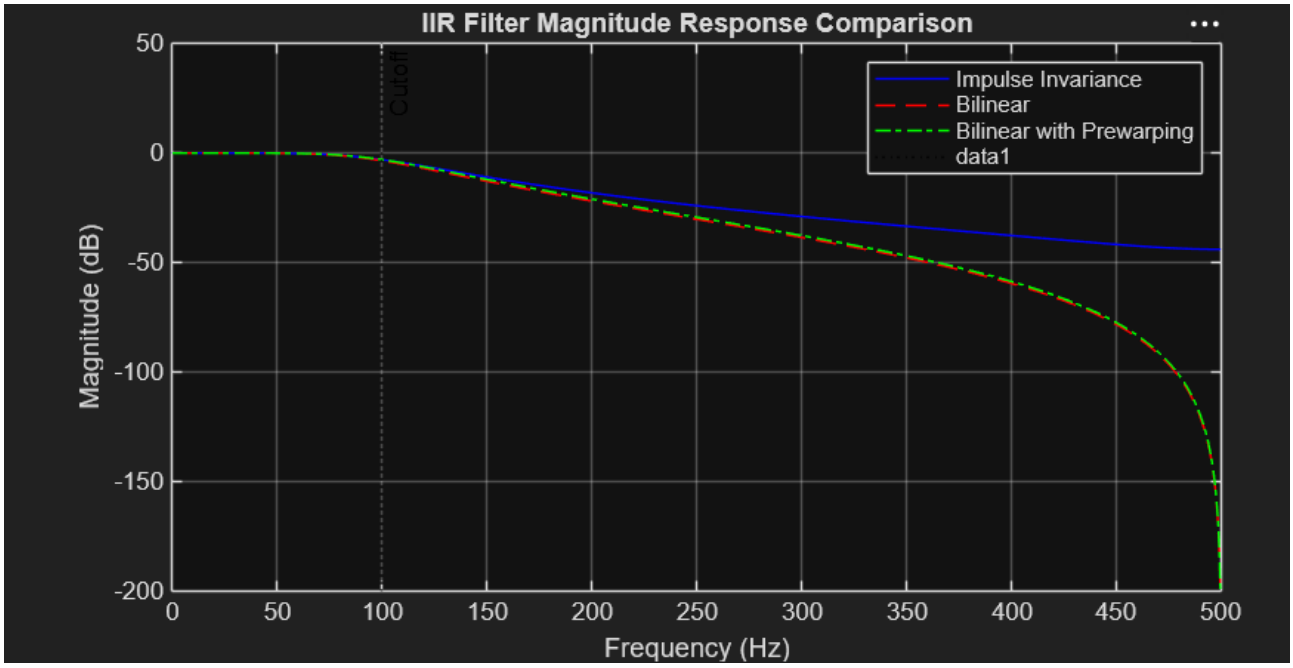
3. Bilinear Transformation with Prewarping
   - Numerator coefficients: `[0.0181, 0.0543, 0.0543, 0.0181]`
   - Denominator coefficients: `[1, -1.7600, 1.1829, -0.2781]`

Design Parameters

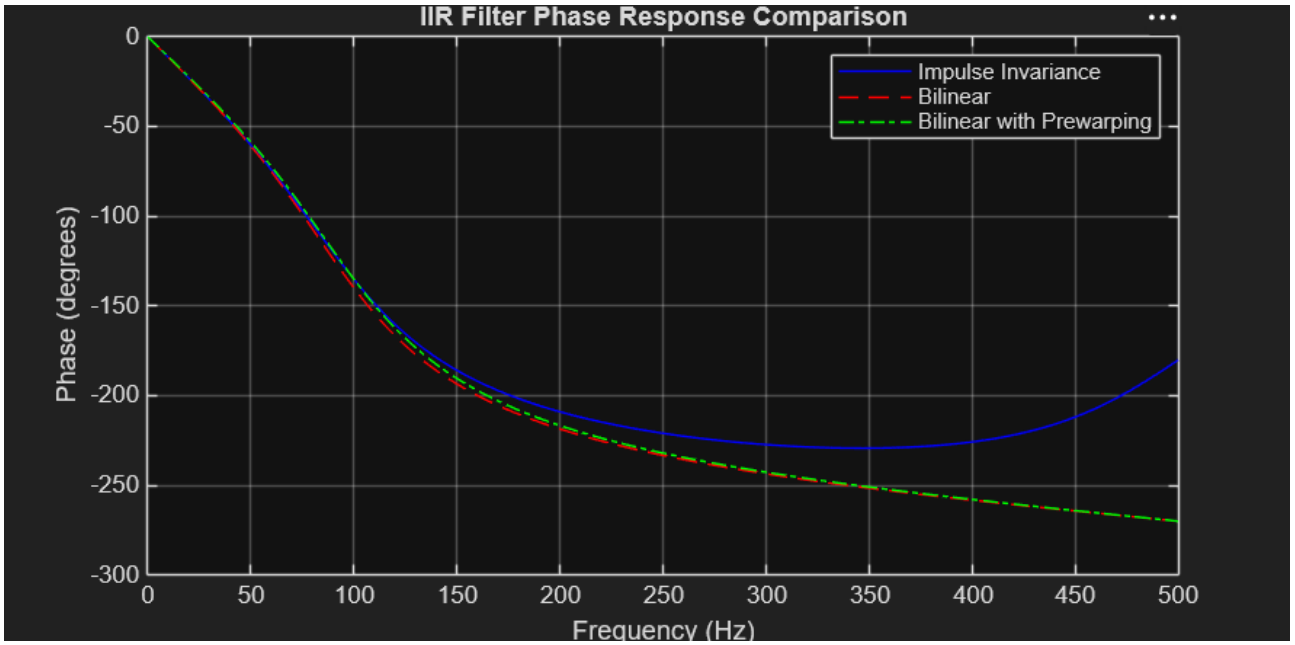
- Filter order: 3
- Sampling frequency: 1000 Hz
- Desired cutoff frequency: 100 Hz
- Prewarped analog cutoff frequency: 649.84 rad/s

Graphs:

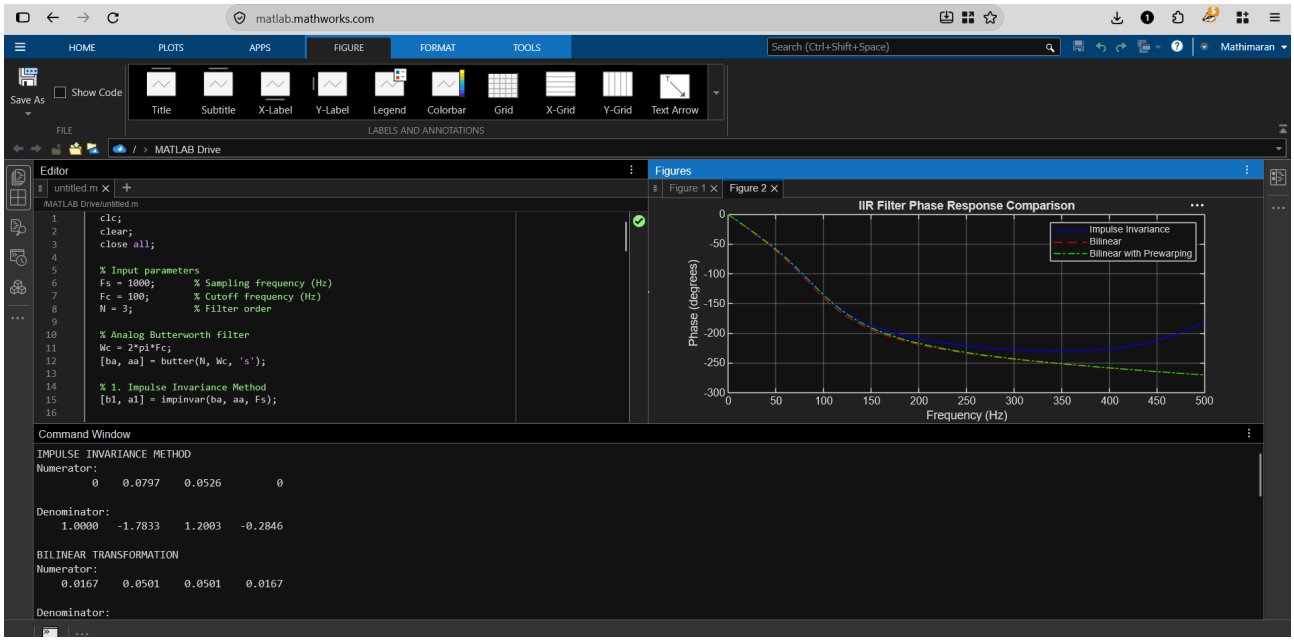
1. Magnitude response graph



2. Phase response graph



Screenshot:



Objectives:

1. To design digital IIR filters using the impulse invariance method in MATLAB.
2. To design digital IIR filters using the bilinear transformation method.
3. To study frequency warping and the effect of prewarping.
4. To obtain the numerator and denominator coefficients of the designed filters.
5. To plot and compare the magnitude and phase responses of the filters obtained using both methods.

---------All the Objectives are simulated----------